

Section A : Pure Mathematics [40 marks]

- 1 Use the substitution $y = (3-x)u$ to show that the differential equation $(3-x)\frac{dy}{dx} = x - y$ can be reduced to $\frac{du}{dx} = \frac{x}{(3-x)^2}$.

Hence obtain the particular solution for which $x < 3$, given that it passes through the origin. Sketch this particular solution. [7]

- 2 Relative to the origin O , points A and B have position vectors $\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix}$ respectively. The line joining B to the origin is denoted by l . Given that C is a point with position vector $k \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$, where k is a fixed constant, show that C' , the reflection of C in the line l has position vector $\frac{k}{7} \begin{pmatrix} 11 \\ 2 \\ -13 \end{pmatrix}$. [4]

Find the value of k such that BC is perpendicular to OA . [2]

With this value of k , verify that $6\overrightarrow{OA} + 15\overrightarrow{OC} = 21\overrightarrow{OC'}$. Hence show that A , C and C' are collinear. [2]

- 3 A complex number w is such that $ww^* - 16\sqrt{3}i + 8iw = 0$ and $\text{Im}(w) < 5$, where w^* is the conjugate of w .

(i) Find w in the form $x + yi$, where $x, y \in \mathbb{R}$. [3]

(ii) Find the integer values of n such that w^n is real. [3]

(iii) Evaluate $1 + \left(\frac{w}{4}\right)^3 + \left(\frac{w}{4}\right)^6 + \left(\frac{w}{4}\right)^9 + \dots + \left(\frac{w}{4}\right)^{21}$. [2]

Another complex number z has modulus 4 and satisfies $\arg\left(\frac{zi}{1+i}\right) = \frac{3}{4}\pi$.

(iv) Express z in the form of $a + bi$, where $a, b \in \mathbb{R}$. [2]

(v) Find the area of the triangle ZWO where Z and W are points on the Argand diagram that represent the complex numbers z and w respectively, and O is the origin. [2]

- 4 Given that $f(x) = e^{\sin x}$, use standard series to find the series expansion for $f(x)$, up to and including the term in x^4 .

Deduce the series expansion of $e^{\sin(x+\pi)}$, up to and including the term in x^4 . [5]

The function $y = g(x)$ satisfies $4 \frac{dy}{dx} = (y+1)^2$, and $y = 1$ at $x = 0$.

- (i) By further differentiation, find the series expansion for $g(x)$, up to and including the term in x^4 . Hence show that when x is small, $g(x) - f(x) \approx \frac{x^3 + x^4}{4}$. [5]
- (ii) By using the result in (i), describe the relationship between the two graphs of f and g when x is close to 0. [1]
- (iii) An approximation to $\int_{-k}^k f(x) dx$ can be found by using $\int_{-k}^k g(x) dx$ where k is a small positive constant. Show that an estimate of the error, E , of this approximation is $\frac{k^5}{10}$. Hence find the range of values of k for E to be less than 10^{-2} . [2]

Section B : Statistics [60 marks]

- 5 (i) Mary is studying the distribution of emergency telephone boxes in a large rural area. Based on a recent finding, it is found that there is an average of 300 emergency telephone boxes per 500 km^2 . She then took a map of a part of this area and divided the area into 50 squares, each of area 1 km^2 . Suggest a suitable probability distribution model that Mary could use for the number of emergency telephone boxes per square. [2]
- (ii) John discovered that the map used by Mary was not updated as it did not reflect the new developments, which included a large city situated within this rural area. John then used an updated map of a part of this area and divided the area into squares of area 1 km^2 . Comment on the suitability of the model suggested in (i) for the number of emergency telephone boxes per square in this situation. [2]
- 6 The governor of a city forms a committee to find out how the recent economic crisis has affected the spending power of the residents. The committee randomly selects a popular shopping mall in the city and obtains a sample of 50 shoppers to participate in a survey.
- (a) One way to obtain the above sample is to randomly choose a shopper from the 1st to 10th shopper who visits the shopping mall, and choose every 10th shopper thereafter until 50 shoppers are obtained. State whether such a sample is obtained using systematic sampling. Justify your answer. [2]
- (b) Describe how another sample of 50 shoppers can be obtained using quota sampling. [2]
- (c) Suppose quota sampling is to be used, suggest two ways to make the sample more representative of the population. [2]

- 7 20 males and 20 females enroll in a dancing course. Of these 40 people, 10 males and 8 females know how to dance hip-hop. Of the 10 males who know how to dance hip-hop, 2 are married. Of the 8 females who know how to dance hip-hop, 4 are married. The rest of the people are not married. 2 males and 2 females are chosen to perform an item. Events A and B are defined as follows:

A : At least 3 chosen people know how to dance hip-hop.

B : All 4 chosen people are married.

Find $P(A)$ and $P(B | A)$. [5]

Determine whether A and B are independent events. Justify your answer. [2]

- 8 In a promotional campaign organized by the management of the petrol filling station, a customer who refills his car with at least \$50 worth of petrol is entitled to draw 5 tickets from a lucky draw box at random. $\frac{1}{6}$ of the tickets in the lucky draw box are winning tickets while the rest are non-winning tickets. The 5 tickets drawn by a customer are put back into the box before the draw by the next customer.

State an assumption needed for a binomial distribution to be a valid model for the number of winning tickets drawn in a draw of 5 tickets. [1]

Using a binomial distribution as the valid model, find

- (i) the probability that among 3 randomly chosen customers who participated exactly once in a draw, only one of them obtained at least one winning ticket in the draw of 5 tickets. [3]
- (ii) using a suitable approximation, the probability that among 60 randomly chosen customers who participated exactly once in the draw, the average number of winning tickets among them is at least one. [2]

Without performing any further calculations, explain why the probability that each of these 60 randomly chosen customers will obtain at least one winning ticket is less than the probability obtained in (ii). [1]

- 9 In a game show, the host will give an **incorrect** arrangement of the letters of a word and let the contestant guess the word within a time limit of one minute. If the word is '**DISTINCTION**', in how many different ways can the host arrange the letters if
- (i) the first and last letters are vowels? [2]
 - (ii) the letters D, I, S, T must not be arranged together in the same order as they appear in the word "DISTINCTION"? [2]
 - (iii) there are exactly 3 letters between the 2 'T's? [3]

- 10** The table below shows the Certificates of Entitlement Quota Premiums for eight bidding exercises in 2009:

Small cars premium x (in thousands of dollars)	4.890	5.116	7.090	7.589	8.489	9.889	11.690	12.899
Big cars premium y (in thousands of dollars)	5.101	5.001	7.501	7.490	7.552	9.180	11.889	14.840

- (i) Obtain the value of the linear (product moment) correlation coefficient r for the data. Explain whether we can conclude that the rise in the big car premiums is due to the rise in the small car premiums. [2]
- (ii) Plot a scatter diagram for the data and explain how its shape is related to the value of r obtained in (i). [2]
- (iii) A motorist wishes to use $y = ax + b$, $y = ab^x$ or $y = ax^b$ to establish a model for the relationship between x and y . Identify the most suitable model and justify your answer. [2]
- (iv) Obtain the values of a and b for the model chosen in (iii). [2]

- 11** The mean SAT score of students in a school for the Mathematics section is known to be m . An education researcher tried a new teaching pedagogy on a random sample of 100 students in this school. The SAT score of these 100 students after the implementation of the new pedagogy, x , gave the following data:

$$\Sigma(x - 120) = 48\,005 \quad \text{and} \quad \Sigma x^2 = 36\,990\,985$$

- (a) A hypothesis test is carried out at 5% significance level and it is found that the new pedagogy has been effective. Find the range of values of m . [4]
- (b) After the implementation of another teaching pedagogy, the SAT score of 10 students are collected. A hypothesis test is carried out at $\alpha\%$ significance level and a p -value is obtained to investigate the effectiveness of this pedagogy.
- (i) State what is meant by ' p -value' in the context of this question. [1]
- (ii) Explain why a t -test instead of a z -test should be carried out. State two assumptions for the test to be valid. [3]
- (iii) If the null hypothesis is not rejected when a t -test is carried out, explain whether it is necessarily true that the same conclusion is obtained when a z -test is carried out at the same significance level. [2]

- 12** In a typical 5-day work week (from Monday to Friday), Andy leaves home for work at 7.00 a.m. He travels to the pick-up point where buses A and B are scheduled to pick him up to his workplace. The time needed for him to travel to the pick-up point follows a normal distribution with a mean of 18 minutes and a standard deviation of 4 minutes.

The schedule for the pick-up time for the buses A and B is as follows:

Bus	Time of departure from the pick-up point
A	7.15 a.m.
B	7.25 a.m.

Andy takes the earliest departing bus when he arrives at the pick-up point. Assume that the time needed for him to board the bus is negligible.

- (i) Show that the probability that Andy takes bus B to his work place is 0.733, correct to 3 significant figures. [1]
- (ii) By using a suitable approximation, find the probability that Andy takes either buses on at most 36 occasions in eight consecutive work weeks. [4]
- (iii) Find the probability that the average time taken for Andy to travel to the pick-up point for the first two days of the week exceeds the average time taken for the last three days of the same week by less than 5 minutes. [3]
- (iv) Andy intends to leave his home earlier so that the probability of him taking bus A is at least 0.90. Find the latest time that he should leave his home? Give your answer correct to the nearest minute. [3]

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